

Does the Source of Carrier Image Affect Steganographic Detectability?

Midway Project Proposal

Project 2.2, Department of Advanced Computing Sciences

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1. Motivation and Problem Statement

Image steganography is an important topic in online security and privacy, with both beneficial and adversarial uses [1, 2]. In practice, detectability depends not only on the embedding method but also on the underlying statistics of the carrier image [9, 21]. Steganalysis works by detecting subtle distributional perturbations in pixel values, residuals, and frequency coefficients after embedding. This is exactly why carrier selection matters: two carriers with identical payload and identical embedding can produce different detectability outcomes.

Most existing steganalysis pipelines were developed and benchmarked on real camera photographs, where sensor noise and compression traces are relatively well characterized. This assumption is increasingly fragile. Photo-realistic synthetic images from modern text-conditional diffusion generators (e.g., Stable Diffusion variants) are now widespread [5], and they are produced by different mechanisms than camera pipelines. Prior synthetic-image forensics confirms that diffusion-generated images carry distinct statistical traces independent of any payload [15], which may directly influence steganographic detectability.

The core question is therefore: **do steganalysis detectors validated on photographs remain equally effective on ML-generated carriers under identical embedding settings?** This matters for three reasons:

- **Security:** if ML-generated carriers are harder to detect, an attacker may evade existing detectors by changing carrier source only.
- **Scientific:** the interaction between generative-model distributions and embedding perturbations remains underexplored.
- **Practical:** as AI-generated images become common in communication channels, the steganographic attack surface expands.

The problem statement for this project is to quantify whether *carrier source category* (real vs. ML-generated, and model-to-model within ML) is associated with measurable detection differences when embedding, payload, and detector settings are held fixed. Because the synthetic images are regenerated from captions rather than matched at the pixel level, we frame the study as a domain-shift comparison rather than a source-only causal test. If source effects are large, detector calibration and risk assumptions built on photo-only datasets may not transfer to mixed real/synthetic traffic.

The closest precedents are Gao et al. [8] and Méreur et al. [25] (see Section 6 for detailed comparison); neither provides a matched real-vs-ML comparison using standardized classical LSB/DCT pipelines and AUC-based steganalysis as the primary endpoint, leaving comparative detectability under those controlled conditions unclear.

This study addresses that gap with a matched design spanning 3 carrier sources $\times$ 2 embedding methods $\times$ 3 payload levels $\times$ 2 encryption conditions over 1,500 carrier images (500 real, 500 ML-A, 500 ML-B), evaluated by training-free statistical detectors matched to each embedding branch. By holding embedding, payload, and detector settings fixed within this comparison, we test whether carrier domain and generator choice are associated with meaningful detection-performance differences. Because training-free detectors have no learned domain bias, any observed AUC gaps are attributable to carrier statistics rather than detector-training mismatch. Results will guide benchmarking and steganalysis policy in mixed real/synthetic environments.

Contributions. (1) A prompt-matched, 3-way carrier dataset (real, SDXL, PixArt- α) purpose-built for steganographic evaluation. (2) A controlled experimental protocol that isolates carrier-domain effects from embedding, payload, and detector variables. (3) An AUC-based confirmatory comparison (DeLong + Bonferroni-Holm) of training-free steganalysis performance on photographic versus diffusion-generated carriers across both spatial and frequency embedding branches.

2. Research Questions

RQ1: Primary (Carrier Source, Real vs. ML)

Does carrier source (real photographs vs. ML-generated images) significantly affect steganographic detectability under matched embedding settings? ML-A and ML-B are pooled into a single ML group to test whether ML origin *in general* changes detectability; RQ2 then decomposes this pool by generator. Because the two generators may differ in opposite directions, pooling could attenuate or inflate the observed effect; we report per-generator breakdowns alongside the pooled result.

RQ2: Primary (Generation-Model Effect Within ML)

Within ML-generated carriers, does the choice of generation model (ML-A vs. ML-B) significantly affect detectability under prompt-matched conditions?

RQ3: Exploratory (Payload Interaction)

Does payload size alter source-related detectability gaps (real vs. ML and ML-A vs. ML-B)?

RQ4: Exploratory (Embedding Branch)

Do the spatial-domain branch (LSB+PNG) and frequency-domain branch (DCT-LSB+JPEG) show different carrier-source effects on detectability? Each branch bundles an embedding algorithm with its natural codec; observed cross-branch differences therefore reflect the complete pipeline rather than embedding algorithm alone.

RQ5: Verification (Encryption Invariance)

Does AES-256-CBC encryption of the payload affect detectability, and does any such effect interact with carrier source or embedding branch? Because AES-256-CBC output is computationally indistinguishable from uniform random bits, embedding statistics presented to the detector should be structurally identical under both conditions, so we expect no effect; this serves as a design-validation check. Any deviation would indicate detector sensitivity to payload structure and would have direct implications for threat modelling.

Verification details for each RQ are specified in Section 4.

3. Chosen Approaches

3.1 Datasets

Real images (500 total). We use **COCO** [16] (300 images) and **Flickr30k** [17] (200 images) for their detailed human-written captions; images with ambiguous captions are excluded. Images are converted to grayscale (luma $0.299R + 0.587G + 0.114B$), resized to 512×512 , and stored as 8-bit PNG (LSB branch) and JPEG Q=95 (DCT branch). Grayscale conversion aligns with the BOSS-Base [19] training corpus and removes color-palette biases of diffusion models from residual statistics.

ML-generated images (1,000 total). From the real set, we generate two prompt-matched sets via the **diffusers** library [24]: **ML-A** (500) with Stable Diffusion XL 1.0 [6] and **ML-B** (500) with PixArt- α [7], comparing a latent U-Net diffusion model against a diffusion-transformer. Each caption is used once per model.

Generation-stage quality control. Generated candidates are screened for two gross failure modes: (1) *saturation failures* (grayscale mean > 245 or < 10 after 512×512 normalization, flagging washed-out or near-black outputs) and (2) *semantic failures* (near-uniform outputs or complete subject absence, assessed on a stratified random 10% holdout per generator model via manual inspection). BRISQUE is excluded: its NSS assumptions [22] do not transfer to diffusion outputs [23]. Images that fail either screen are re-generated from the same caption with a new random seed; if re-generation fails a second time the caption is dropped entirely and replaced by the next unused caption in the dataset, preserving the target $N=500$ per source group.

3.2 Embedding Methods

We implement two canonical steganographic methods:

LSB substitution (spatial domain) replaces the

k least significant bits of each pixel channel value with message bits in row-major order. Three payload levels are defined by the fraction of available positions filled:

Table 1: Payload level definitions. Levels are matched across branches by fill rate (fraction of available embedding positions used). Both branches use $k=1$ LSB replacement; spatial capacity is 1 bpp (one bit per pixel), DCT capacity is 1 bit per non-zero AC coefficient.

Level	Fill	Spatial bpp	DCT fill
Low	25%	0.25	25% of non-zero ACs
Medium	50%	0.50	50% of non-zero ACs
High	75%	0.75	75% of non-zero ACs
<i>Bit-depth sensitivity (reported separately, not in RQ3 analysis):</i>			
BD-Sens	75% ($k=2$)	1.50	—

The three primary payload levels all use $k=1$ to isolate fill-rate effects from bit-depth effects. The BD-Sens condition ($k=2$, 1.50 bpp) is a spatial-branch-only sensitivity check, excluded from the payload-interaction analysis in RQ3 since it changes both bit-depth and fill rate simultaneously.

DCT-LSB embedding (frequency domain) operates on JPEG carriers (Q=95). Following the JSteg approach [10, 13], message bits replace the least significant bit of non-zero quantized AC DCT coefficients in row-major block order. Zero-valued coefficients and DC coefficients are skipped to avoid perceptible distortion and gross histogram artifacts. This mirrors the spatial-branch embedding logic (LSB replacement) but operates in the frequency domain, making both branches structurally parallel while producing different statistical footprints.

Implementation pipeline. DCT-LSB operates directly on the quantized integer DCT coefficients stored in the JPEG bitstream, accessed via **jpegio** (Python), which exposes the coefficient matrix without pixel-domain decoding. The modified coefficients are re-entropy-coded at the same Q=95 table without re-quantization, avoiding double-quantization distortion.

3.3 Encryption

Payloads are embedded under two conditions (plain and AES-256-CBC), applied uniformly across both branches.

3.4 Image-Quality Tests (Quality Control)

For each stego/cover pair we compute PSNR, SSIM, and FSIM [20] to rule out trivial degradation effects.

3.5 Steganalysis Detectors

All primary detectors are training-free statistical tests, each reimplemented in Python from the original publications. Training-free detectors carry no learned domain bias, so any AUC differences across carrier sources reflect the carrier statistics themselves rather than a detector-training mismatch. Because both branches use LSB replacement (on pixel values and on DCT coefficients respectively), the χ^2 attack applies to both, giving each branch at least two independent detectors.

Spatial-branch detectors. RS analysis [9] classifies pixel groups as Regular or Singular under flipping masks; the ratio shift between cover and stego images estimates embedding rate. **The χ^2 attack** [10] tests whether adjacent value pairs ($2k, 2k+1$) converge to equal frequency,

Table 2: Steganalysis detector overview. All primary detectors are training-free.

Detector	Principle	Training	Domain
RS Analysis [9]	Regular/Singular groups	None	Spatial
χ^2 attack [10]	Pairs-of-values frequency	None	Spatial
Sample Pairs [11]	Trace multisets	None	Spatial
χ^2 attack (DCT) [10]	Pairs-of-values on DCT coeff.	None	DCT
Calibration χ^2 [13]	Crop-recompress reference	None	DCT

a signature of LSB replacement. **Sample Pairs (SP) analysis** [11] counts trace multisets from pixel pairs, providing a third independent estimate. All three assume sequential (row-major) embedding order.

DCT-branch detectors. The χ^2 attack (DCT) [10] applies the same pairs-of-values test to quantized DCT coefficients rather than pixel values; since DCT-LSB replaces coefficient LSBs identically to spatial LSB, the test transfers directly. **Calibration χ^2** [13] provides an independent second detector: the stego image is cropped by four pixels at a non-block-aligned offset and re-compressed at Q=95, giving an embedding-free reference; a χ^2 test on the resulting DCT histogram pairs detects embedding-induced shifts.

Extension (time-permitting): SRNet [18], a deep residual network for steganalysis, retrained on BOSS-Base [19] images embedded with spatial LSB and DCT-LSB respectively. If completed, SRNet would add a trained detector to both branches, reported separately from the primary training-free analysis.

3.6 Limitations

With 500 images per source group, the design targets medium-to-large effects; confirmatory testing is restricted to RQ1–2 (Bonferroni–Holm), with RQ3–4 reported by effect size. Generator coverage is limited to two families. Crucially, ML-A (SDXL, latent U-Net) and ML-B (PixArt- α , diffusion transformer) differ not only in architecture but also in training data and conditioning pipeline, so any observed RQ2 difference is architecture-plus-training-data confounded and cannot be attributed to architecture alone. The spatial branch has three detectors while the DCT branch has two; this minor asymmetry reflects the available training-free methods in the literature. Grayscale conversion, while necessary for BOSSBase alignment, discards inter-channel correlations that may carry source-distinguishing information, potentially attenuating the measured carrier-source effect. Cross-branch comparisons (RQ4) conflate embedding algorithm with codec; within-branch results for RQ1–2 are free of this confound.

4. Experiments

Statistical protocol: Exp. 1–2 are **confirmatory** (Bonferroni–Holm, $\alpha = 0.05$); Exp. 3–4 are **exploratory**, reported via AUC differences with 95% confidence intervals; Exp. 5 is a **verification** check (expected null). A *stratum* is defined as the tuple (detector, embedding method, payload level). We use the DeLong test [14] because the same images pass through all detectors, making per-stratum AUC estimates correlated; Bonferroni–Holm is preferred over plain Bonferroni for higher power when

some null hypotheses are false.

Exp. 1 (RQ1): Carrier Source, real vs. pooled ML. Apply all detectors per branch (spatial: RS, χ^2 , SP; DCT: χ^2 , calibration χ^2). For each stratum (detector, method, payload), compare real vs. pooled ML AUC via DeLong test and report differences with 95% CIs.

Exp. 2 (RQ2): Generation-model effect within ML. For each stratum (detector, method, payload), compare AUC between prompt-matched ML-A and ML-B pairs using the DeLong test [14]. Report detector-specific AUC differences with 95% confidence intervals.

Exp. 3 (RQ3): Payload interaction. Plot AUC and source-contrast gaps across Low, Medium, and High payload levels (all $k=1$; LSB bpp: 0.25, 0.50, 0.75; DCT fill: 25%, 50%, 75% of non-zero ACs). Report source $\times$ payload interaction with effect sizes. The BD-Sens condition (1.50 LSB bpp, $k=2$, LSB branch only) is reported separately as a sensitivity check and excluded from this analysis.

Exp. 4 (RQ4): Embedding-branch interaction. Compare carrier-source AUC gaps stratified by branch (spatial LSB+PNG vs. DCT-LSB+JPEG). Each branch bundles embedding with its natural codec, so cross-branch differences reflect the full pipeline rather than the algorithm alone; within-branch results for RQ1–2 allow stronger attribution. Report source $\times$ branch interaction via AUC differences with 95% CIs.

Exp. 5 (RQ5): Encryption invariance check. Compare AUC between plain and AES-256-CBC conditions per source group, branch, and detector. Since AES-256-CBC output is computationally indistinguishable from random bits, encryption should not change detection rates; a null result validates the experimental design. Any deviation would indicate detector sensitivity to payload structure. Report encryption main effect and source $\times$ encryption interaction via AUC differences with 95% CIs.

5. Prototype

Vertical prototype (algorithm depth): We validate the core components separately before scaling: spatial LSB + RS/ χ^2 /SP, and DCT-LSB + χ^2 /calibration χ^2 on a small test subset.

Horizontal prototype (integration breadth): We then run an integrated pipeline on a mixed 60-image subset (20 real + 20 ML-A + 20 ML-B) at medium payload to validate interfaces and outputs before executing the full 1,500-image study.

6. Related Work

6.1 Classical Embedding and Detection Foundations

Image steganography has predominantly focused on embedding secret data into photographic carriers using spatial- and frequency-domain techniques [1, 3]. The canonical example is LSB substitution, which replaces the least significant bit of sample values — pixel intensities in the spatial domain, or quantized DCT coefficients in JPEG (the JSteg approach) [10, 13]. We adopt LSB replacement in both domains rather than introducing a

new embedding algorithm, to preserve comparability with prior steganalysis literature and focus on how detectability changes across carrier domains.

On the detection side, RS analysis [9], the χ^2 attack [10], and Sample Pairs analysis [11] provide independent training-free baselines for LSB-like embedding. Fridrich et al. [13] introduced calibration-based training-free detection for JPEG via re-compression at a non-block-aligned offset. Higher-capacity trained detectors such as SRM [4] and DCTR [12] exist but require reimplementa-tion of high-dimensional feature extraction pipelines; we use the training-free methods listed above, which additionally avoid domain-bias confounds in our cross-source comparison.

6.2 AI-Generated Carriers and Closest Prior Work

Gao et al. [8] demonstrate that generative models can encode steganographic secrets as coverless shares — where the AI-generated image constitutes the secret-bearing object itself rather than a carrier for an embedded payload — establishing that generative models produce images suitable for information concealment; this is a distinct paradigm from the classical payload-embedding studied here. Most closely related, Méreur et al. [25] study cover-source mismatch introduced by a wide range of AI text-to-image generators (diffusion models and GANs) as steganographic carriers, showing that synthetic covers significantly exploit detector mismatch. However, they use adaptive embedding algorithms (HILL, UNIWARD, Mi-POD) rather than classical LSB pipelines, and report total probability of error P_E rather than AUC; their study is also large-scale and exploratory rather than controlled and confirmatory. Mallet et al. [26] provide a systematic review of the broader cover-source mismatch problem. Our study fills a complementary gap: a controlled classical pipeline (spatial and DCT-domain LSB embedding, training-free detectors, AUC+DeLong confirmatory inference) applied to real photographs versus prompt-conditioned outputs from two distinct diffusion architectures.

6.3 Cross-Domain and Synthetic-Image Forensics

Cross-domain steganalysis has typically examined camera-to-camera distribution shifts. Giboulot et al. [21] show that detectors trained on one camera source suffer measurable AUC degradation on another. Our study applies this framing to a larger shift: from camera photographs to latent diffusion outputs, which lack the optical sensor noise and JPEG compression history that steganalysis detectors exploit. Corvi et al. [15] separately confirm that diffusion-generated images carry distinct statistical traces independent of any payload, further supporting our hypothesis.

6.4 Classical vs. Deep Steganalysis

Deep-learning detectors such as SRNet [18] can achieve high benchmark accuracy, but with higher compute requirements and reduced interpretability. We prioritize classical detectors for the primary analysis, consistent with the project’s cryptography/steganography focus. Our contribution is not a new embedding method or detector, but a controlled evaluation of how existing classical steganalysis transfers across carrier domains.

7. Relation to Curriculum

This project connects directly to **Computer Security** (AES-256-CBC and adversarial threat modeling), **AI and Machine Learning** (diffusion-based image generation and cross-domain evaluation), **Software Engineering and Architectures** (modular, testable pipeline design), and **Statistics** (DeLong AUC testing, effect sizes, and confidence intervals for rigorous comparison).

8. Planning

Gantt charts for both Phase 2 (implementation) and Phase 3 (completion and reporting) are provided in Appendix A.

9. Minimal Passing Requirements

The study must implement one encryption scheme, two embedding approaches (spatial and frequency domain), and domain-appropriate detectors, and must provide defensible answers to RQ1 (carrier-source effect: real vs. ML) and RQ2 (generation-model effect within ML).

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A. Project Gantt Charts

Phase 2: Implementation

Period 5: 30 March – 15 May 2026 (7 weeks)

WBS	Task	Start	End	Days	Wk 1	Wk 2	Wk 3	Wk 4	Wk 5	Wk 6	Wk 7
1	Data Construction										
1.1	Dataset collection (COCO/Flickr30k with captions)	30.03	10.04	10							
1.2	ML image generation (SDXL + PixArt- α)	30.03	10.04	10							
1.3	Heuristic quality screening & dual-format normalisation	07.04	17.04	8							
2	Steganography Implementation										
2.1	LSB embedding pipeline (sequential, $k=1, 2$, PNG)	07.04	17.04	8							
2.2	DCT-LSB embedding pipeline (JPEG Q=95)	07.04	17.04	8							
2.3	AES-256-CBC payload encryption	14.04	17.04	4							
3	Detection & Analysis										
3.1	RS Analysis + χ^2 detector	14.04	24.04	8							
3.2	Sample Pairs detector & calibration χ^2 setup	14.04	01.05	14							
3.3	Image quality metrics (PSNR/SSIM/FSIM)	21.04	01.05	8							
3.4	Encryption-effect experiments (RQ5)	28.04	08.05	8							
3.5	Statistical analysis & effect-size reporting	28.04	08.05	8							
4	Writing & Revision										
4.1	Visualisations + report writing	04.05	15.05	8							
4.2	Buffer / final revision	11.05	15.05	5							

■ Data ■ Steganography ■ Detection ■ Evaluation ■ Analysis ■ Writing

Milestones: **M1** (end Wk 2): Dataset ready **M2** (end Wk 3): Embedding complete **M3** (end Wk 5): Detection + metrics complete **M4** (end Wk 7): Final report submitted

Phase 3: Completion

Project Period 6: 25 May – 12 June 2026 (3 weeks)

WBS	Task	Start	End	Days	Wk 1	Wk 2	Wk 3
1.1	Complete remaining implementation	25.05	05.06	10			
1.2	Verify results & rerun experiments	25.05	05.06	10			
1.3	Statistical verification & effect sizes	01.06	05.06	5			
2.1	Presentation slides	01.06	08.06	6			
2.2	Poster	05.06	10.06	4			
2.3	Final paper write-up	01.06	12.06	10			
2.4	Buffer / revision / submission	10.06	12.06	3			

■ Implementation ■ Verification ■ Analysis ■ Slides ■ Poster ■ Paper

Milestones: **M5** (end Wk 1): Implementation finalised **M6** (end Wk 2): Results verified, slides ready **M7** (end Wk 3): All deliverables submitted